NETWORK DISCOVERY REPORT
NETWORK SCANNING WITH ZENMAP

W2-PM5 | WEEK 2 – PROJECT MODULE 5
CYBERSECURITY & ETHICAL HACKING

Project	Network Scanning with Zenmap (Network Discovery Phase)
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Program	Networkwalks | Batch B084
Report date	11 November 2026
Environment	Windows PC – home Wi-Fi LAN
Target subnet	192.168.1.0/24
Assessment type	Authorized local network discovery and limited service scan

Prepared for educational and portfolio purposes. Findings are limited to the screenshots and scan outputs captured for this project.

1. Liability Disclaimer

This activity was conducted for cybersecurity education and portfolio documentation on a home Wi-Fi network, with the knowledge of the network owner. Network scanning must only be performed on systems and devices that the tester owns or is explicitly authorized to assess. Do not scan networks belonging to other people or organizations without prior permission. These point-in-time observations do not establish that a device is vulnerable or compromised.

2. Introduction

This report documents W2-PM5: Network Scanning with Zenmap. Zenmap is the graphical interface for Nmap and makes network discovery and scanning tasks accessible through a GUI. The Networkwalks module requires the learner to install Zenmap, identify the local IP address and subnet, discover live hosts, record host IP and MAC addresses, and save the network topology as a PDF.

3. Objectives

- Identify the Windows PC's local IPv4 address and LAN subnet.
- Discover responding hosts on the local /24 network using a Ping Scan.
- Record the IP addresses and hostnames of the live hosts.
- Identify MAC address information where the captured evidence provides it.
- Review the network topology and limited TCP service-scan output.

4. Scope and Methodology

In scope: local subnet 192.168.1.0/24 and the devices that responded during the recorded scans.

Tools: Zenmap/Nmap on Windows and Windows Command Prompt (`ipconfig /all`).

Discovery method: Nmap Ping Scan (`nmap -sn 192.168.1.11/24`). Nmap normalizes the target 192.168.1.11/24 to the network 192.168.1.0/24 (256 addresses).

Service scan: Zenmap Quick Scan profile (`nmap -T4 -F 192.168.1.11/24`), which checks the 100 most common TCP ports of each live host.

Limitations: the number of responding hosts varied between scans; Nmap reported an Npcap warning and fell back to connect() mode, so MAC addresses of other hosts were not captured; the Quick Scan output for the last host (192.168.1.13) was cut off in the screenshot; the topology PDF export was not supplied.

5. Tools Used

Tool	Purpose
Zenmap (Nmap GUI)	GUI for running Nmap host-discovery and port-scanning commands.
Nmap Ping Scan (-sn)	Checks which IP addresses respond as live hosts; it is not a port scan.
Nmap Quick Scan (-T4 -F)	Fast scan of the 100 most common TCP ports on each live host.
Windows CMD – ipconfig /all	Displays adapter configuration, IPv4 address, subnet mask, gateway and physical (MAC) address.
Windows Wi-Fi adapter	Realtek RTL8852BE WiFi 6 802.11ax adapter used by the PC during the scans.

6. Activities Performed and Results

6.1 Task 1 – Download and Install Zenmap

Nmap/Zenmap was downloaded from the official Nmap download page and installed on the Windows PC. The screenshots confirm that Zenmap launched and ran scans successfully.

Official download page: https://nmap.org/download.html

6.2 Task 2 – Identify Local IP Address and LAN Subnet

The `ipconfig /all` output (Figure A) shows the active Wireless LAN adapter with IPv4 address **192.168.1.11** and subnet mask **255.255.255.0**, which corresponds to subnet **192.168.1.0/24**. The default gateway and DHCP server are both 192.168.1.1.

6.3 Task 3 – Discover Live Hosts

The primary Zenmap Ping Scan (Figure B) reports **7 hosts up** out of 256 IP addresses scanned. The live hosts are:

No.	Hostname shown	IPv4 address
1	Unnamed (router / gateway)	192.168.1.1
2	Ayesha-s-A36	192.168.1.4
3	Galaxy-A57-5G	192.168.1.5
4	Uni.*20DVR (DVR)	192.168.1.6
5	Redmi-Note-9-Pro-Max	192.168.1.7
6	SHAIK (local PC)	192.168.1.11
7	V2336	192.168.1.13

Scan history. Three scans were captured on 11 November 2026, in this order according to the Nmap timestamps: a Ping Scan at 12:06 that found 5 hosts (Figure F), a Quick Scan at 12:11 that listed 7 hosts (Figure D), and the Ping Scan at 12:26 that found 7 hosts (Figure B). Hosts 192.168.1.5 and 192.168.1.7 were missing from the first scan, which is normal for mobile devices that sleep, roam or join the Wi-Fi later. The 7-host result is the most complete and is used for Tasks 3–5. Nmap's timestamps differ from the Windows taskbar clock (06:22 PM), so they are used here only to show the order of the scans.

Note on the Zenmap host list. The host panel in Figure B also lists an iPhone (192.168.1.8). Zenmap keeps hosts from earlier scans in its list, but this host is absent from the Nmap output of that scan ("7 hosts up"), so it is not counted as a live host for this scan.

6.4 Task 4 – Number of Live Hosts

7 live hosts, including the local PC. The Nmap output states: "256 IP addresses (7 hosts up) scanned in 18.57 seconds".

6.5 Task 5 – IP Addresses of Live Hosts

192.168.1.1, 192.168.1.4, 192.168.1.5, 192.168.1.6, 192.168.1.7, 192.168.1.11 and 192.168.1.13.

6.6 Task 6 – MAC Addresses of Live Hosts

The `ipconfig /all` output identifies the Physical Address of the local Wi-Fi adapter as **CC-47-40-F5-7E-B4** for the PC at 192.168.1.11. Nmap displays the MAC addresses of other LAN hosts only when it can use raw packets. In this run it reported that it could not import all Npcap functions and fell back to connect() mode, so no MAC Address lines appear for the other hosts. Their MAC addresses cannot be confirmed from the captured evidence and are not guessed.

Host	IPv4 address	MAC address
SHAIK (local PC)	192.168.1.11	CC-47-40-F5-7E-B4 (confirmed)
Other 6 live hosts	192.168.1.1, .4, .5, .6, .7, .13	Not captured (see Section 9)

6.7 Task 7 – Network Topology

The Zenmap topology (Figure C) shows the scanning machine as the central *localhost* node, with the other hosts connected around it. Under Zenmap's legend, a green node is a host with fewer than 3 open ports and a yellow node is a host with 3 to 6 open ports. This matches the Quick Scan: the router (192.168.1.1, 3 open ports) and the PC (192.168.1.11, 4 open ports) are yellow, and the DVR (2 open ports) and the remaining hosts are green. The topology also shows the iPhone (192.168.1.8) from an earlier scan, which is not in the 7-host list. The task asks for the topology to be saved as a PDF (Topology tab → Save Graphic → PDF); the exported PDF was not supplied, so Figure C is included as the available evidence.

7. Additional Port-Scan Observations

The Quick Scan (`nmap -T4 -F`) checked the 100 most common TCP ports of each host. An open port is not automatically a vulnerability; further validation is required before assigning severity.

Host	Open TCP ports	Service labels shown
192.168.1.1 (router)	53, 80, 443	domain (DNS), HTTP, HTTPS
192.168.1.6 (DVR)	80, 554	HTTP, RTSP
192.168.1.11 (local PC)	135, 139, 445, 6646	MSRPC, NetBIOS-SSN, Microsoft-DS, unknown

Hosts 192.168.1.4, 192.168.1.5 and 192.168.1.7 showed all 100 scanned ports closed or in ignored states. The result for 192.168.1.13 is cut off in the screenshot and is therefore not reported. The service on 6646/tcp was labelled “unknown” and should be identified (for example with `netstat -ano` on the PC or a version scan with `nmap -sV` on this authorized host).

8. Risk Analysis / Impact

Observation	Potential impact	Priority
Multiple live devices on one LAN	An attacker already on the same network could map the connected devices.	Informational
Router exposes DNS, HTTP and HTTPS	Management/service interfaces may be reachable; risk depends on configuration and authentication.	Review configuration
DVR exposes HTTP and RTSP	Video-streaming or administration services should be protected from unauthorized access.	Review configuration
Windows PC exposes SMB/NetBIOS-related ports	File sharing and legacy services increase the attack surface if enabled unnecessarily.	Review configuration
Unknown service on 6646/tcp (PC)	An unidentified listening service cannot be assessed until it is identified.	Investigate
MAC addresses of other hosts not captured	Asset identification is incomplete in the captured evidence.	Evidence gap

9. Security Recommendations

- Change default router and DVR administrator credentials; use unique, strong passwords.
- Disable remote administration and unused services where not required.
- Restrict DVR/RTSP access to trusted devices or networks.
- Keep the router, DVR and Windows systems updated.
- Disable SMB/NetBIOS or file sharing when not needed; restrict access with Windows Firewall.
- Identify the process listening on 6646/tcp and disable it if it is not required.
- Use WPA2-AES or WPA3 Wi-Fi security with a strong passphrase; consider a separate guest/IoT network.
- Repeat the inventory scan periodically and maintain an authorized asset list.
- To capture MAC addresses, install the latest Npcap from https://npcap.com, run Zenmap as Administrator and repeat the Ping Scan; the Windows command `arp -a` can also list the IP-to-MAC mappings of hosts on the local network.

10. Conclusion

The W2-PM5 exercise demonstrated local network discovery using Zenmap. The Windows adapter configuration identified 192.168.1.11/24, and the primary Ping Scan found 7 live hosts on 192.168.1.0/24. The MAC address of the local adapter was confirmed using `ipconfig /all`. The Quick Scan recorded open TCP services on the router, the DVR and the local Windows PC, including an unidentified service on 6646/tcp. MAC addresses of the remaining hosts and the exported topology PDF were not part of the captured evidence and should be added once they are collected.

11. Evidence Register

Evidence file	What it demonstrates
CMD.png	Figure A: Windows ipconfig /all – local IPv4 address, subnet mask, gateway and Wi-Fi MAC.
Screenshot 2026-11-11 182235.png	Figure B: Primary Zenmap Ping Scan (12:26) – 7 hosts up.
topology.png	Figure C: Topology diagram with device labels.
quick scan .png	Figure D: Quick Scan (12:11) – open TCP services of selected hosts.
ports.png	Figure E: Zenmap Ports/Hosts view of the router.
list.png	Figure F: Earlier Ping Scan (12:06) – 5 hosts up.
zenmap.png	Zenmap GUI screenshot and discovered host list.

12. References

1. Networkwalks, *W2-PM5 – Week 2 Project Module 5: Network Scanning with Zenmap* (practical guide).
2. Nmap official download page: https://nmap.org/download.html
3. Nmap Reference Guide: https://nmap.org/book/man.html
4. Npcap: https://npcap.com

Appendix – Selected Screenshots

```
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
   Physical Address. . . . . . . . . : CC-47-40-F5-7E-B4
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2401:4900:1c0f:5f0c:356e:5a6a:e91f:74ca(Preferred)
   Temporary IPv6 Address. . . . . . : 2401:4900:1c0f:5f0c:d867:7e7c:9c9e:6d96(Preferred)
   Link-local IPv6 Address . . . . . : fe80::3a28:9e7c:4385:fdd5%17(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.1.11(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 10 November 2026 19:31:47
   Lease Expires . . . . . . . . . . : 12 November 2026 11:13:56
   Default Gateway . . . . . . . . . : fe80::3215:77ff:fee9:9f40%17
                                       192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.1
   DHCPv6 IAID . . . . . . . . . . . : 181159744
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-32-43-7C-DA-CC-47-40-F5-7E-B4
   DNS Servers . . . . . . . . . . . : fe80::3215:77ff:fee9:9f40%17
                                       192.168.1.1
                                       fe80::3215:77ff:fee9:9f40%17
   Primary WINS Server . . . . . . . : 192.168.1.1
   NetBIOS over Tcpip. . . . . . . . : Enabled

C:\Users\shaik>
```

Figure A. Windows ipconfig /all – local adapter details and MAC.

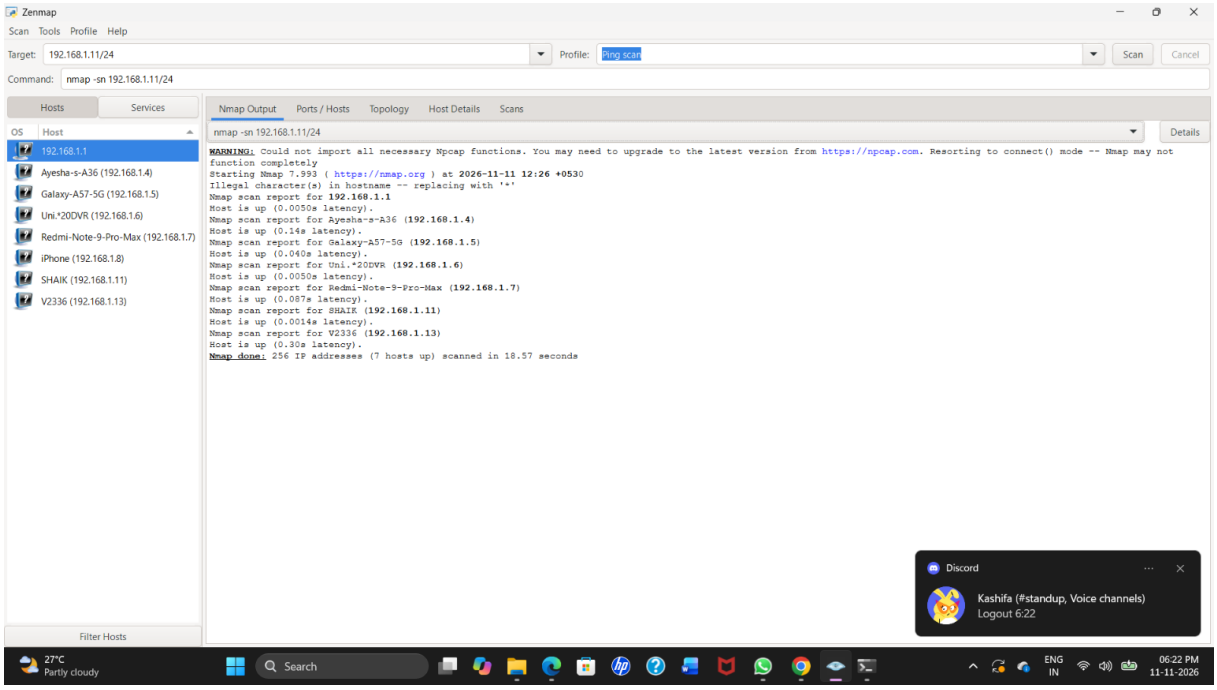


Figure B. Primary Zenmap Ping Scan – 7 hosts up (12:26).

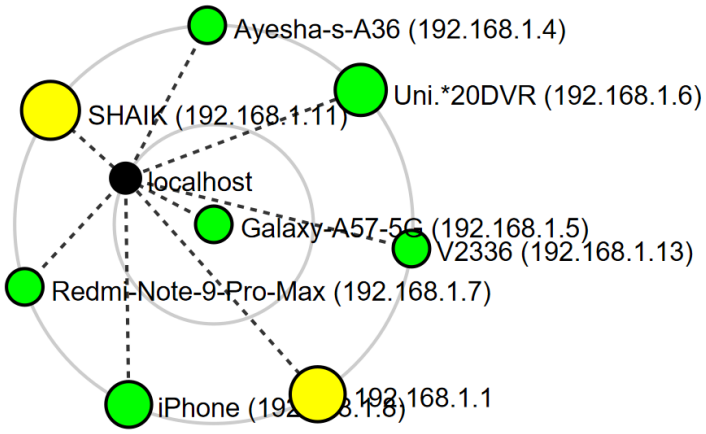


Figure C. Zenmap topology.

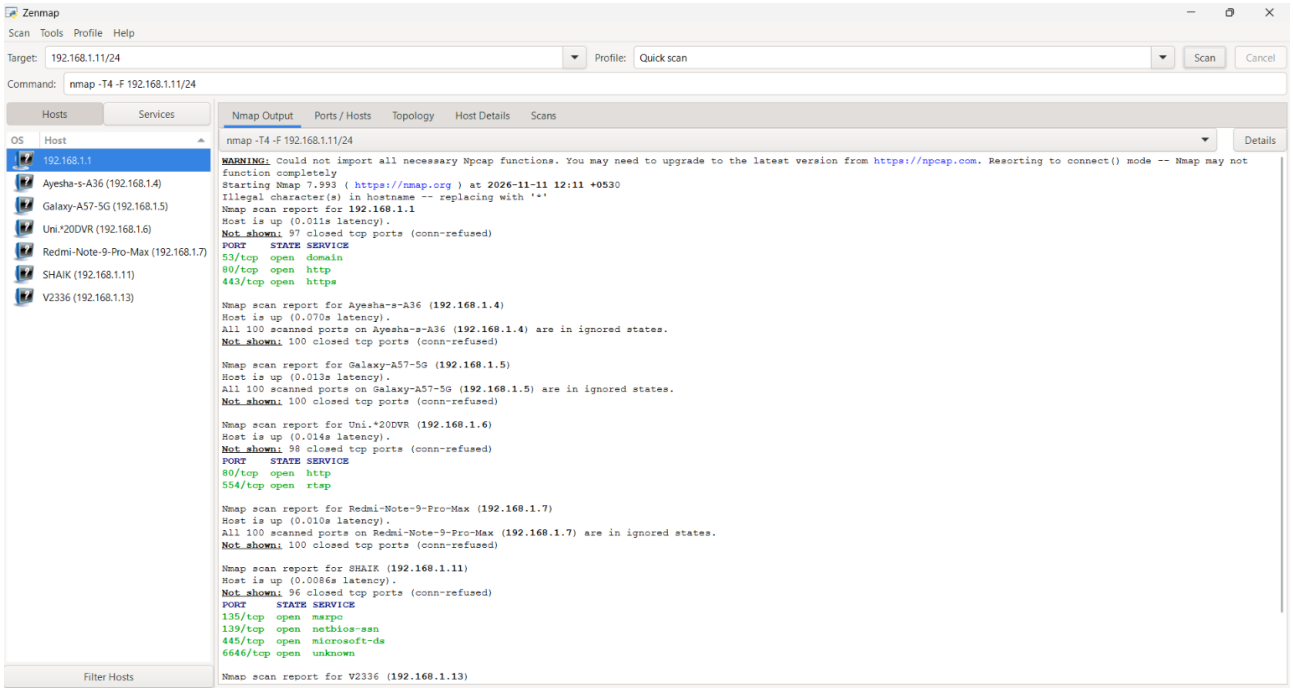


Figure D. Zenmap Quick Scan output (12:11).

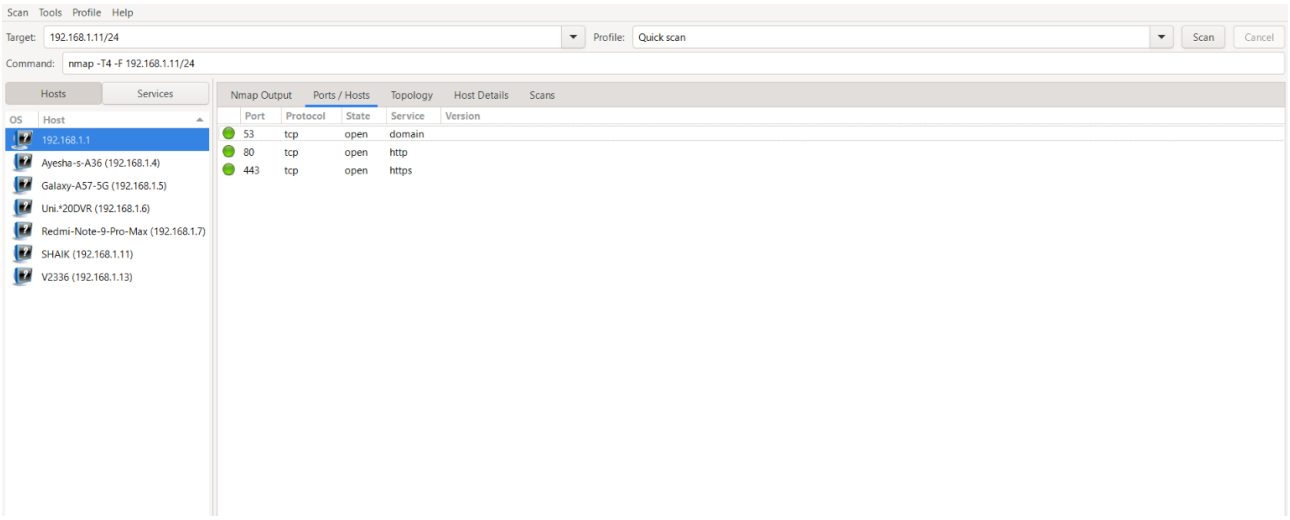


Figure E. Ports/Hosts view of the router (192.168.1.1).

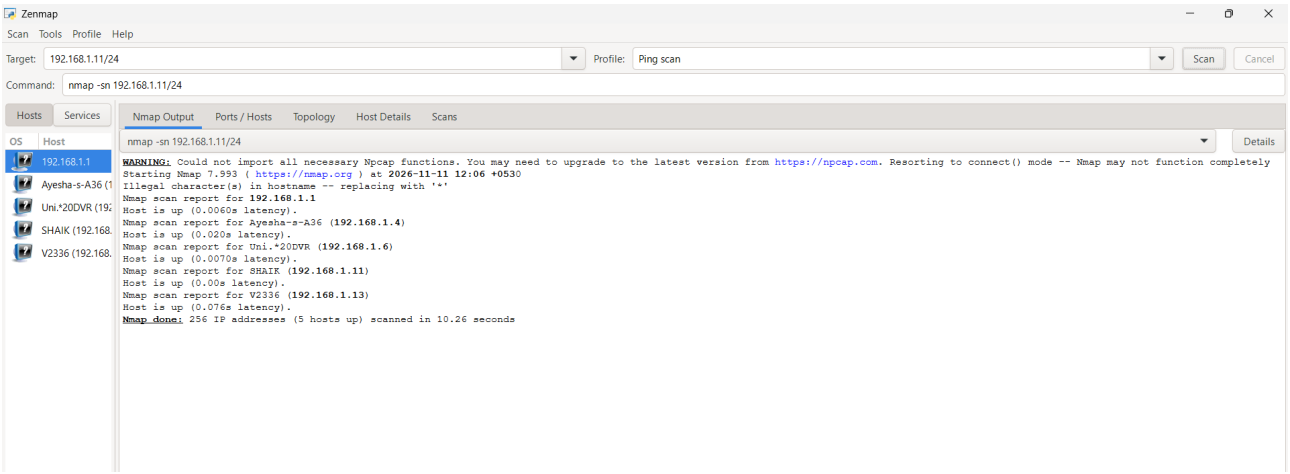


Figure F. Earlier Ping Scan – 5 hosts up (12:06).